DTC P252F:00 [SKYACTIV-D 1.5]

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Except European (L.H.D., U.K.) specs.

DTC P252F:00	Engine oil level too high
DETECTION CONDITION	• The amount of engine oil dilution **exceeds 1,661 g {58.59 oz}** during the diesel particulate filter regeneration control. **Diagnostic support note** • This is a continuous monitor (other). • The check engine light does not illuminate. • FREEZE FRAME DATA/Snapshot data is not available. • DTC is stored in the PCM memory.
FAIL-SAFE FUNCTION	Not applicable
POSSIBLE CAUSE	• Engine oil attenuation by frequently diesel particulate filter regeneration • Engine oil data reset procedure does not perform in each engine oil replacements. • PCM malfunction
SYSTEM WIRING DIAGRAM	Not applicable

Diagnostic Procedure

STEP	INSPECTION		ACTION
1	**RECORD VEHICLE STATUS WHEN DTC WAS DETECTED TO UTILIZE WITH REPEATABILITY VERIFICATION** **Note** • Recording can be facilitated using the screen capture of the PC function. • Record the freeze frame data/snap shot data.	—	Go to the next step.
2	**VERIFY RELATED SERVICE INFORMATION AVAILABILITY** • Verify related Service Information availability. • Is any related Service Information available?	Yes	Perform repair or diagnosis according to the available Service Information. • If the vehicle is not repaired, go to the next step.
		No	Go to the next step.
3	**INSPECT ENGINE OIL LEVEL** • Inspect the engine oil level. (See ENGINE OIL LEVEL INSPECTION [SKYACTIV-D 1.5].) • Is the engine oil amount above X line on the dipstick?	Yes	Replace the engine oil. (See ENGINE OIL REPLACEMENT [SKYACTIV-D 1.5].) Perform the “ENGINE OIL DATA RESET”, then go to Step 5. (See ENGINE OIL DATA RESET [SKYACTIV-D 1.5].)
		No	Go to the next step.
4	**VERIFY RELATED PENDING CODE AND/OR DTC** • Switch the ignition off, then ON (engine off). • Perform the Pending Trouble Code Access Procedure and DTC Reading Procedure. (See ON-BOARD DIAGNOSTIC TEST [SKYACTIV-D 1.5].) • Is the PENDING CODE/DTC P253F:00 also present?	Yes	Replace the engine oil. (See ENGINE OIL REPLACEMENT [SKYACTIV-D 1.5].) Perform the “ENGINE OIL DATA RESET”, then go to the next step. (See ENGINE OIL DATA RESET [SKYACTIV-D 1.5].)
		No	Go to the next step.

STEP	INSPECTION		ACTION
5	**VERIFY DTC TROUBLESHOOTING COMPLETED** • Clear the DTC from the PCM memory using the M-MDS. (See AFTER REPAIR PROCEDURE [SKYACTIV-D 1.5].) • Start the engine and warm it up completely. **Caution** **• While performing this step, always operate the vehicle in a safe and lawful manner.** **• When the M-MDS is used to observe monitor system status while driving, be sure to have another technician with you, or record the data in the M-MDS using the PID/DATA MONITOR AND RECORD capturing function and inspect later.** • Drive the vehicle under the FREEZE FRAME DATA/snapshot data condition. • Perform the DTC Reading Procedure. (See ON-BOARD DIAGNOSTIC TEST [SKYACTIV-D 1.5].) • Is the same DTC present?	Yes	Replace the PCM, then go to the next step. (See PCM REMOVAL/INSTALLATION [SKYACTIV-D 1.5].)
		No	Go to the next step.
6	**VERIFY AFTER REPAIR PROCEDURE** • Perform the “AFTER REPAIR PROCEDURE”. (See AFTER REPAIR PROCEDURE [SKYACTIV-D 1.5].) • Are any DTCs present?	Yes	Go to the applicable DTC inspection. (See DTC TABLE [SKYACTIV-D 1.5].)
		No	DTC troubleshooting completed.

Note

- To determine the malfunctioning part, proceed with the diagnostics from “Function Inspection Using M-MDS”.

Details On DTCs (European (L.H.D., U.K.) specs.)

DESCRIPTION	Engine oil level signal: engine oil level high	
DETECTION CONDITION	Determination conditions	• Oil level increased above or near X mark.
	Preconditions	• Engine speed: **750—3,000 rpm** • Vehicle speed: **5 km/h {4 mph} or more** • Engine oil control: during low oil pressure • Engine oil temperature: **20—120 °C {68—248 °F}** • The following DTC is not detected: — Communication error between engine oil level sensor and PCM: U1100:00 — Engine oil level sensor malfunction: P250A:00 — Engine oil level sensor power supply circuit: P250C:00
	Drive cycle	• **1**
	Self test type	• CMDTC self test
	Sensor used	• PCM • Engine oil level sensor • Engine oil temperature sensor • Low-G sensor
FAIL-SAFE FUNCTION	• Not applicable	
VEHICLE STATUS WHEN DTCs ARE OUTPUT	• Wrench indicator light is illuminated	

DESCRIPTION	Engine oil level signal: engine oil level high
POSSIBLE CAUSE	• Improper engine oil level (high) • Engine oil attenuation by frequently diesel particulate filter regeneration • Engine oil level sensor malfunction — Clogging of oil passage to engine oil level sensor detection area — Engine oil level sensor internal application specific integrated circuit malfunction • Erratic signal to PCM — Engine oil level sensor connector or terminals malfunction — Open or short circuit in wiring harness between engine oil level sensor terminal C and PCM terminal 1Q • PCM malfunction

System Wiring Diagram
- Not applicable

Function Explanation (DTC Detection Outline)
- The PCM monitors the oil level while the above preconditions are being met and if the traveled distance is **100 km {62.2 mph} or more** or the monitoring time is **300 s or more**, it calculates the average oil level.
- If the averaged oil level is near the X mark two times in a row, the PCM stores a DTC and turns on the wrench indicator light in the instrument cluster.

Repeatability Verification Procedure
1. Move the vehicle to a level area and switch the ignition ON.
2. Verify engine oil level using the dipstick (verify that it is near the X mark).

PID Item/Simulation Item Used In Diagnosis

PID/DATA monitor item table

Item	Definition	Unit	Condition/Specification
EOL	Engine oil level	mm, in	• Displays the engine oil level.

Function Inspection Using M-MDS

STEP	INSPECTION		ACTION
1	**PURPOSE: VERIFY RELATED SERVICE INFORMATION AVAILABILITY** • Verify related Service Information availability. • Is any related Service Information available?	Yes	Perform repair or diagnosis according to the available Service Information. • If the vehicle is not repaired, go to the next step.
		No	Go to the next step.
2	**PURPOSE: IDENTIFY TRIGGER DTC FOR FREEZE FRAME DATA** • Is the DTC P252F:00 on FREEZE FRAME DATA?	Yes	Go to the next step.
		No	Go to the troubleshooting procedure for DTC on FREEZE FRAME DATA. (See DTC TABLE [SKYACTIV-D 1.5].)
3	**PURPOSE: RECORD FREEZE FRAME DATA/ SNAPSHOT DATA AND DIAGNOSTIC MONITORING TEST RESULTS TO UTILIZE WITH REPEATABILITY VERIFICATION** **Note** • Recording can be facilitated using the screen capture function of the PC. • Record the FREEZE FRAME DATA/snapshot data and DIAGNOSTIC MONITORING TEST RESULTS (misfire related) on the repair order.	—	Go to the next step.
4	**PURPOSE: VERIFY IF DIAGNOSTIC RESULT IS AFFECTED BY OTHER RELATED DTCs OCCURRING** • Switch the ignition off, then ON (engine off). • Perform the Pending Trouble Code Access Procedure and DTC Reading Procedure. (See ON-BOARD DIAGNOSTIC TEST [SKYACTIV-D 1.5].) • Is the other PENDING CODE/DTC also present?	Yes	Go to the applicable DTC inspection. (See DTC TABLE [SKYACTIV-D 1.5].)
		No	Go to the next step.

STEP	INSPECTION		ACTION
5	**PURPOSE: VERIFY IF THERE IS PID ITEM CAUSING DRASTIC CHANGES OF ACCELERATION FLUCTUATION BY INPUT SIGNAL TO PCM** • Access the following PIDs using the M-MDS: (See ON-BOARD DIAGNOSTIC TEST [SKYACTIV-D 1.5].) **PCM:** — EOL • Is there any signal that is far out of specification?	Yes	Go to the next step.
		No	Go to the troubleshooting procedure to perform the procedure from Step 1.
6	**PURPOSE: VERIFY CONNECTOR CONNECTIONS** • Access the following PIDs using the M-MDS: (See ON-BOARD DIAGNOSTIC TEST [SKYACTIV-D 1.5].) **PCM:** — EOL • When the following parts are shaken, does the PID value include a PID item which has changed? — Engine oil level sensor — PCM	Yes	Repair or replace the applicable connector parts. Go to the troubleshooting procedure to perform the procedure from Step 3.
		No	Go to the troubleshooting procedure to perform the procedure from Step 1.

Troubleshooting Diagnostic Procedure
Intention of troubleshooting procedure

- Step 1
  - — Perform inspection of engine oil level.
- Step 2
  - — Inspect for clogging of the oil passage to the engine oil level sensor detection area.
- Step 3—4
  - — Verify that the primary malfunction is resolved and there are no other malfunctions.

STEP	INSPECTION		ACTION
1	**PURPOSE: INSPECT ENGINE OIL LEVEL** • Inspect the engine oil level. (See ENGINE OIL LEVEL INSPECTION [SKYACTIV-D 1.5].) • Is the engine oil amount above X line on the dipstick?	Yes	Replace the engine oil. (See ENGINE OIL REPLACEMENT [SKYACTIV-D 1.5].) Perform the “ENGINE OIL DATA RESET”, then go to Step 3. (See ENGINE OIL DATA RESET [SKYACTIV-D 1.5].)
		No	Go to the next step.
2	**PURPOSE: INSPECT FOR CLOGGING OF THE OIL PASSAGE TO THE ENGINE OIL LEVEL SENSOR DETECTION AREA** • Remove the engine oil level sensor. (See ENGINE OIL LEVEL SENSOR REMOVAL/ INSTALLATION [SKYACTIV-D 1.5].) • Visually inspect the oil passage to the engine oil level sensor. • Is there any restriction detected?	Yes	Clean the oil passage, then go to the next step.
		No	Replace the engine oil level sensor, then go to the next step. (See ENGINE OIL LEVEL SENSOR REMOVAL/ INSTALLATION [SKYACTIV-D 1.5].)
3	**PURPOSE: PERFORM DTC INSPECTION AND VERIFY IF MALFUNCTIONING PART IS PCM** • Always reconnect all disconnected connectors. • Clear the DTC from the PCM memory using the M-MDS. (See CLEARING DTC [SKYACTIV-D 1.5].) • Implement the repeatability verification procedure. (See Repeatability Verification Procedure.) • Perform the DTC Reading Procedure. (See ON-BOARD DIAGNOSTIC TEST [SKYACTIV-D 1.5].) • Is the same DTC present?	Yes	Repeat the inspection from Step 1. • If the malfunction recurs, replace the PCM. (See PCM REMOVAL/INSTALLATION [SKYACTIV-D 1.5].) Go to the next step.
		No	Go to the next step.

STEP	INSPECTION		ACTION
4	**VERIFY AFTER REPAIR PROCEDURE** • Perform the “AFTER REPAIR PROCEDURE”. (See AFTER REPAIR PROCEDURE [SKYACTIV-D 1.5].) • Are any DTCs present?	Yes	Go to the applicable DTC inspection. (See DTC TABLE [SKYACTIV-D 1.5].)
		No	DTC troubleshooting completed.